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4 TODO GUI **6**

1 DONE

1.1 DONE

1. ☒ UI
2. ☒
3. ☒
4. ☒
5. ☒ AST
6. ☒

1.2 DONE

1. ☒
2. ☒ I/O

2 DONE

2.1 DONE

1. ☒
2. ☒
3. ☒
4. ☒ ,

2.2 DONE trimmer

1. ☒ -> .

2.3 DONE

1. ☒ next
2. ☒ next
3. ☒ enum
4. ☒
5. ☒
6. ☒ str
7. ☒ -> (,)

2.4 DONE

1. ☒ ast node
2. ☒ struct node tree as vec
3. ☒ parser match the syntax of the node and build it to the tree -> match in progres
   - ☒ each child node is degree of progress -> no need to store in parser struct
   - ☒ match node type in case of (none, none), (left, none), (left, right)->go to parent - = ->

(a) ☒ Program,
(b) ☒ FunctionList,
(c) ☒ Function
(d) ☒ FunctionInterface,
(e) ☒ ParameterList,
(f) ☒ DeclarationList,
(g) ☒ Declaration,
(h) ☒ Declarator,
(i) ☒ ValueKind,
(j) ☒ StatementList,
(k) ☒ Statement(StatementKind),
    ☒ Expression
    ☒ If
    ☒ While -> If
    ☒ Compound
    ☒ Return,
    ☒ Empty
(l) ☒ AssginExpr,
(m) ☒ CondtionExpr,
(n) ☒ AdditiveExpr(AdditiveOpKind),
    ☒ add
    ☒ sub
(o) ☒ Term(MultiplicativeOpKind),
(p) ☒ UnaryExpr(UnaryOpKind),
(q) ☒ Factor(FactorKind),
(r) ☒ Identifier(String),
(s) ☒ Literal(i32),

2.5 DONE

1.

☒ (,)

☒

☒

☒ ,

☒ ,

☒ ,

2.

☒ . .

☒ int .

☒ , .

☒ (LValue) (RValue) .

☒ , , RValue .

☒ LValue . ().

☒ RValue . Lvalue .

☒ Lvalue Rvalue .

☒ .

☒ .

☒ , , .

3.

☒ .

☒ .

☒ .

☒ if, while .

☒ .

☒ .

☒ return .

☒

2.6 DONE IR

1. I/O

   - PUT, GET
   - ->

2. , , , , ,

   - -> , , .

- -> , .
- . ->
- .
- -> . 1 .
- -> . immediate load .
- -> .
- -> ,

3.

- (add,sub,mul,div) . immediate x
- store, load, loadi .
- cmpeq( ), cbr() .
- jumpi .

2.7 DONE

1.

- ->
- ->
-
- Cmpeq -> 1 0

2.

- .

3. ,

- .
- .
- .
  – , SP ,
- .
- .
- caller, callee .

- , .
  - PC BSA . -> =
  - BUN .
  - BUN .
  - BSA.
- , 2 . + 2

4. , , , ,
   - , , .
   - . ->
   - .

5.
   -
   - .

6. -

2.8 DONE

2.9 DONE

3 DONE

4 TODO GUI